

# TABLE OF CONTENTS

|                                                                          |          |
|--------------------------------------------------------------------------|----------|
| Executive Summary . . . . .                                              | iii      |
| List of Tables . . . . .                                                 | v        |
| List of Figures . . . . .                                                | vi       |
| List of Abbreviations . . . . .                                          | vii      |
| List of Symbols . . . . .                                                | viii     |
| <br>                                                                     |          |
| <b>1 Introduction</b>                                                    | <b>1</b> |
| 1.1 Background . . . . .                                                 | 1        |
| 1.2 Motivation . . . . .                                                 | 2        |
| 1.3 Scope of the Project . . . . .                                       | 2        |
| <br>                                                                     |          |
| <b>2 Project Description and Goals</b>                                   | <b>4</b> |
| 2.1 Literature Review . . . . .                                          | 4        |
| 2.1.1 Object Detection for Road Hazards . . . . .                        | 4        |
| 2.1.2 Convolutional Neural Networks and Residual Architectures . . . . . | 5        |
| 2.1.3 Infrastructure Monitoring Systems . . . . .                        | 5        |
| 2.1.4 Quantum Machine Learning . . . . .                                 | 5        |
| 2.1.5 Hybrid Quantum-Classical Models . . . . .                          | 6        |
| 2.2 Research Gap . . . . .                                               | 6        |
| 2.3 Problem Statement . . . . .                                          | 7        |
| 2.4 Objectives . . . . .                                                 | 7        |
| 2.5 Summary . . . . .                                                    | 7        |
| <br>                                                                     |          |
| <b>3 Technical Specification</b>                                         | <b>8</b> |
| 3.1 Requirements . . . . .                                               | 8        |
| 3.1.1 Functional Requirements . . . . .                                  | 8        |
| 3.1.2 Non-Functional Requirements . . . . .                              | 8        |
| 3.2 Feasibility Study . . . . .                                          | 9        |
| 3.2.1 Technical Feasibility . . . . .                                    | 9        |
| 3.2.2 Economic Feasibility . . . . .                                     | 9        |

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| 3.2.3    | Operational Feasibility . . . . .              | 9         |
| 3.3      | System Specification . . . . .                 | 9         |
| 3.3.1    | Hardware Requirements . . . . .                | 9         |
| 3.3.2    | Software Requirements . . . . .                | 10        |
| 3.4      | Summary . . . . .                              | 10        |
| <b>4</b> | <b>System Design</b>                           | <b>11</b> |
| 4.1      | System Architecture . . . . .                  | 11        |
| 4.2      | Detailed Design . . . . .                      | 12        |
| 4.2.1    | Input and Detection Module . . . . .           | 12        |
| 4.2.2    | Preprocessing Pipeline . . . . .               | 12        |
| 4.2.3    | ResNet-18 Backbone . . . . .                   | 13        |
| 4.2.4    | MLP Dimensionality Reduction . . . . .         | 13        |
| 4.2.5    | Variational Quantum Circuit . . . . .          | 13        |
| 4.2.6    | Classification Layer . . . . .                 | 14        |
| 4.2.7    | System Flow . . . . .                          | 14        |
| 4.3      | Summary . . . . .                              | 15        |
| <b>5</b> | <b>Methodology and Testing</b>                 | <b>16</b> |
| 5.1      | Methodology . . . . .                          | 16        |
| 5.1.1    | Stage 1: Detection . . . . .                   | 16        |
| 5.1.2    | Stage 2: Hybrid Classification Model . . . . . | 16        |
| 5.2      | Algorithm . . . . .                            | 18        |
| 5.3      | Testing Procedure . . . . .                    | 18        |
| 5.3.1    | Training Configuration . . . . .               | 18        |
| 5.3.2    | Loss Function . . . . .                        | 18        |
| 5.3.3    | Quantum Gradient Computation . . . . .         | 18        |
| 5.3.4    | Evaluation Metrics . . . . .                   | 19        |
| 5.4      | Summary . . . . .                              | 19        |
| <b>6</b> | <b>Project Implementation</b>                  | <b>20</b> |
| 6.1      | Implementation Overview . . . . .              | 20        |
| 6.2      | Development Environment . . . . .              | 20        |
| 6.2.1    | Software Tools . . . . .                       | 20        |

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| 6.2.2    | Hardware Setup . . . . .                          | 20        |
| 6.3      | Training Configuration . . . . .                  | 21        |
| 6.3.1    | Loss Function . . . . .                           | 21        |
| 6.3.2    | Optimisation . . . . .                            | 21        |
| 6.3.3    | Quantum Gradient Computation . . . . .            | 21        |
| 6.4      | Hyperparameters . . . . .                         | 21        |
| 6.5      | Experimental Configurations . . . . .             | 21        |
| 6.6      | Evaluation Metrics . . . . .                      | 22        |
| 6.6.1    | Accuracy . . . . .                                | 22        |
| 6.6.2    | Precision . . . . .                               | 22        |
| 6.6.3    | Recall . . . . .                                  | 22        |
| 6.6.4    | F1-Score . . . . .                                | 22        |
| 6.6.5    | Confusion Matrix . . . . .                        | 23        |
| 6.6.6    | mAP@50 . . . . .                                  | 23        |
| 6.7      | Baseline Model . . . . .                          | 23        |
| 6.8      | Summary . . . . .                                 | 23        |
| <b>7</b> | <b>Results and Discussion</b>                     | <b>24</b> |
| 7.1      | Introduction . . . . .                            | 24        |
| 7.2      | Overall Classification Performance . . . . .      | 24        |
| 7.2.1    | Analysis of Experimental Configurations . . . . . | 24        |
| 7.3      | Per-Class Analysis . . . . .                      | 25        |
| 7.3.1    | Per-Class Discussion . . . . .                    | 25        |
| 7.4      | Qualitative Pipeline Visualisation . . . . .      | 25        |
| 7.5      | Intermediate Feature Representations . . . . .    | 25        |
| 7.5.1    | ResNet Features . . . . .                         | 25        |
| 7.5.2    | MLP Latent Space . . . . .                        | 26        |
| 7.5.3    | Quantum Measurements . . . . .                    | 27        |
| 7.6      | Training Convergence Analysis . . . . .           | 29        |
| 7.6.1    | Observations . . . . .                            | 29        |
| 7.7      | Discussion . . . . .                              | 29        |
| 7.7.1    | Hybrid Model Effectiveness . . . . .              | 29        |
| 7.7.2    | Role of VQC . . . . .                             | 29        |

|          |                                            |           |
|----------|--------------------------------------------|-----------|
| 7.7.3    | Limitations . . . . .                      | 29        |
| 7.8      | Summary . . . . .                          | 30        |
| <b>8</b> | <b>Conclusion and Further Enhancements</b> | <b>31</b> |
| 8.1      | Conclusion . . . . .                       | 31        |
| 8.1.1    | Summary of Contributions . . . . .         | 31        |
| 8.1.2    | Key Findings . . . . .                     | 32        |
| 8.1.3    | Significance of the Work . . . . .         | 32        |
| 8.2      | Limitations . . . . .                      | 32        |
| 8.3      | Further Enhancements . . . . .             | 33        |
| 8.3.1    | Deployment on Quantum Hardware . . . . .   | 33        |
| 8.3.2    | Noise-Robust Circuit Design . . . . .      | 33        |
| 8.3.3    | Scaling Qubit Count . . . . .              | 33        |
| 8.3.4    | Severity-Based Classification . . . . .    | 33        |
| 8.3.5    | Real-Time System Integration . . . . .     | 34        |
| 8.3.6    | Larger and Diverse Datasets . . . . .      | 34        |
| 8.3.7    | Quantum Kernel Methods . . . . .           | 34        |
| 8.3.8    | Explainability . . . . .                   | 34        |
| 8.4      | Closing Remarks . . . . .                  | 34        |
|          | <b>Appendix A: Source Code</b>             | <b>37</b> |
|          | <b>Appendix B: Publication</b>             | <b>41</b> |

## EXECUTIVE SUMMARY

Urban road infrastructure is critical to the functioning of modern smart cities, yet road surfaces are subject to continuous degradation caused by vehicular loading, weathering, and deferred maintenance. Road hazards such as potholes, surface cracks, and open or displaced manhole covers represent significant threats to public safety and incur substantial economic costs for municipal authorities worldwide. The conventional approach to road inspection — manual field surveys — is labour-intensive, geographically incomplete, and inherently subjective. Advances in computer vision and deep learning have created an opportunity to automate the detection and classification of road hazards at scale.

This thesis presents the design, implementation, and evaluation of a novel **Quantum-Enhanced Deep Learning Framework for Road Hazard Safety Analysis**. The proposed system introduces a two-stage, modular pipeline that deliberately decouples the tasks of hazard localisation and hazard classification. In the detection stage, a YOLO-based object detector is employed to identify and localise regions of interest within full-resolution road images. In the classification stage, the extracted hazard crops are processed by a hybrid quantum-classical classifier that integrates a pre-trained ResNet-18 convolutional backbone, a dimensionality-reducing multilayer perceptron (MLP), and a 4-qubit Variational Quantum Circuit (VQC).

The ResNet-18 backbone, pre-trained on the ImageNet dataset, extracts a rich 512-dimensional feature vector from each  $64 \times 64$  cropped input image using residual learning. A two-layer MLP then compresses this high-dimensional representation to a compact 4-dimensional latent vector, matching the dimensionality required for quantum encoding. The VQC encodes each latent dimension onto a dedicated qubit via parameterised  $R_Y$  rotation gates (angle encoding), applies six variational layers with trainable single-qubit rotations and ring-topology CNOT entanglement gates, and produces output measurements as Pauli-Z expectation values. A classical linear classification head maps these quantum measurements to class logits, and a softmax function yields the final class probability distribution over four hazard categories: cracks, potholes, open manholes, and garbage.

The system is trained on a custom class-balanced dataset constructed by merging

two publicly available annotated road hazard repositories, totalling 6,797 images across four hazard classes. To address class imbalance, random downsampling of the majority class is applied, resulting in a final training set of 3,340 images with approximately uniform class representation. All quantum gradients are computed using the parameter-shift rule, enabling seamless end-to-end backpropagation through the hybrid model.

Experimental evaluation demonstrates that the proposed hybrid model (Classical Dataset + Quantum Model configuration) achieves a macro-averaged F1-score of 0.8439, a precision of 0.8579, and a recall of 0.8451, which is competitive with the purely classical baseline ( $F1 = 0.8016$ ). Per-class analysis shows strong performance across all hazard categories, with potholes achieving an F1-score of 0.92, cracks 0.82, garbage 0.88, and open manholes 0.79. Training convergence analysis confirms that the integration of the VQC does not introduce optimisation instability.

This work constitutes a rigorous proof-of-concept for quantum-augmented infrastructure monitoring and contributes to the emerging field of quantum machine learning applied to real-world safety applications. Future directions include deployment on physical NISQ quantum processors, noise-robust variational ansatz design, and integration with real-time urban monitoring platforms.

## LIST OF TABLES

|     |                                                                    |    |
|-----|--------------------------------------------------------------------|----|
| 6.1 | Training Hyperparameters . . . . .                                 | 22 |
| 7.1 | Performance Comparison Across Dataset and Model Configurations . . | 24 |
| 7.2 | Per-Class Metrics — Proposed Hybrid Model (Test Set) . . . . .     | 25 |

## LIST OF FIGURES

|     |                                                                        |    |
|-----|------------------------------------------------------------------------|----|
| 4.1 | Overall architecture of the proposed quantum-classical framework . . . | 11 |
| 7.1 | Sample input image from the test set . . . . .                         | 26 |
| 7.2 | YOLO-based cropped hazard region . . . . .                             | 26 |
| 7.3 | Final output of the pipeline with predicted class . . . . .            | 27 |
| 7.4 | ResNet-18 feature representations . . . . .                            | 27 |
| 7.5 | MLP representation . . . . .                                           | 28 |
| 7.6 | VQC measurement outputs across classes . . . . .                       | 28 |
| 7.7 | Training and validation accuracy (left) and loss (right) . . . . .     | 29 |

## LIST OF ABBREVIATIONS

|               |                                  |
|---------------|----------------------------------|
| <b>CNN</b>    | Convolutional Neural Network     |
| <b>IoU</b>    | Intersection over Union          |
| <b>mAP</b>    | Mean Average Precision           |
| <b>MLP</b>    | Multilayer Perceptron            |
| <b>NISQ</b>   | Noisy Intermediate-Scale Quantum |
| <b>QML</b>    | Quantum Machine Learning         |
| <b>ReLU</b>   | Rectified Linear Unit            |
| <b>ResNet</b> | Residual Network                 |
| <b>UAV</b>    | Unmanned Aerial Vehicle          |
| <b>ViT</b>    | Vision Transformer               |
| <b>VQC</b>    | Variational Quantum Circuit      |
| <b>YOLO</b>   | You Only Look Once               |

## SYMBOLS AND NOTATIONS

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| $\alpha$                          | Learning rate                        |
| $\theta$                          | Trainable quantum circuit parameters |
| $\mathbf{e} \in [-1, 1]^4$        | Quantum measurement output vector    |
| $\mathbf{f} \in \mathbb{R}^{512}$ | Feature vector from ResNet           |
| $\mathbf{z} \in \mathbb{R}^4$     | Latent vector for quantum encoding   |
| $\mathcal{L}_{\text{CE}}$         | Cross-entropy loss                   |
| $L$                               | Number of VQC layers                 |
| $n$                               | Number of qubits                     |
| $R_Y(\theta)$                     | Single-qubit rotation gate           |
| $Z_k$                             | Pauli-Z operator on qubit $k$        |
| $ \psi\rangle$                    | Quantum state vector                 |

# CHAPTER 1

## INTRODUCTION

### 1.1 BACKGROUND

Urban road networks form a fundamental part of modern infrastructure, supporting the movement of people, goods, and services in everyday life. As cities continue to grow and traffic density increases, maintaining the structural condition of road surfaces becomes increasingly important, not only for transportation efficiency but also for public safety. However, road surfaces naturally degrade over time due to factors such as repeated vehicular loading, temperature variations, moisture, and delayed maintenance. This deterioration typically appears in the form of potholes, surface cracks, and exposed or displaced manhole covers.

These road hazards are a serious issue in both developed and developing regions. Potholes, for instance, can cause tyre damage, suspension issues, and misalignment, leading to significant repair costs. Cracks, if not addressed early, tend to expand and develop into larger structural failures, increasing maintenance costs over time. Open or displaced manholes pose immediate risks to vehicles and pedestrians. In addition to physical damage, such hazards can contribute to accidents and also increase the financial burden on municipal authorities.

Traditionally, road inspection is carried out through manual surveys conducted by trained personnel. These inspections may be done on foot or using vehicles equipped with specialised instruments. While this approach can provide accurate assessments, it is difficult to scale across large road networks. Cities often have thousands of kilometres of roads, making frequent and complete inspections impractical. Moreover, manual inspections can vary depending on the individual performing them, and lower-priority roads may not be inspected regularly.

With the increasing availability of high-resolution cameras, UAVs, and vehicle-mounted sensing systems, there is now a large amount of visual data available for automated analysis. These systems continuously capture road surface images under different conditions. However, converting this raw data into useful insights requires automated systems that can reliably detect and classify different types of road hazards

under varying lighting, weather, and environmental conditions.

## 1.2 MOTIVATION

Although deep learning has shown strong performance in road hazard detection, several challenges still remain:

1. **Intra-class Variability:** Hazards within the same category can look very different. For example, potholes vary in size and shape, and cracks can appear in multiple forms such as longitudinal, transverse, or irregular patterns. This variation makes classification difficult, especially with limited training data.
2. **Inter-class Similarity:** Some hazard types share similar visual features. For instance, shadows may resemble cracks, and debris may partially hide potholes. This makes it harder for models to clearly distinguish between classes.
3. **Data Scarcity and Imbalance:** Compared to large-scale datasets, road hazard datasets are relatively small. Additionally, some classes appear more frequently than others, leading to imbalance and biased model predictions.
4. **Generalisation Across Environments:** Models trained on one dataset may not perform well in different environments due to changes in lighting, road materials, or camera setups.
5. **Limitations of Classical Models:** Classical deep learning models may struggle when feature representations are reduced to very low dimensions, as required in hybrid quantum systems.

These challenges motivate the need for improved approaches that can effectively capture complex visual patterns while maintaining robustness across different environments.

## 1.3 SCOPE OF THE PROJECT

This work focuses on the detection and classification of four types of road hazards: cracks, potholes, open or displaced manholes, and garbage. A YOLO-based detector is

used to identify regions of interest during inference, while the main contribution lies in the design of a hybrid quantum-classical classification model.

The proposed system follows a two-stage pipeline, where detection and classification are handled separately. The classification stage combines a pre-trained ResNet-18 model for feature extraction, a multilayer perceptron for dimensionality reduction, and a 4-qubit Variational Quantum Circuit for enhanced representation learning.

The quantum circuit is implemented using the PennyLane simulator, as execution on physical quantum hardware is beyond the scope of this project. However, the circuit design uses standard quantum gates such as  $R_Y$  rotations and CNOT operations, making it compatible with current quantum computing platforms.

The dataset used in this work is constructed by merging multiple publicly available sources and applying preprocessing techniques such as cropping, resizing, and normalisation. The system is evaluated using standard performance metrics including accuracy, precision, recall, F1-score, and confusion matrix.

The scope of this project is limited to image-based classification and does not include real-time deployment or integration with live monitoring systems.

# CHAPTER 2

## PROJECT DESCRIPTION AND GOALS

### 2.1 LITERATURE REVIEW

#### 2.1.1 Object Detection for Road Hazards

Real-time object detection has improved significantly in recent years, particularly with the development of single-stage detectors. The YOLO (You Only Look Once) family has become widely used due to its ability to perform detection efficiently in a single forward pass.

Wang et al. (2023) introduced YOLOv7, which achieved strong performance in both speed and accuracy on benchmark datasets. The model incorporates architectural improvements such as re-parameterised convolutions and improved training strategies, making it suitable for real-time applications like road hazard detection.

Later, Wang et al. (2024) proposed YOLOv9 with Programmable Gradient Information (PGI), which helps preserve useful information during training. This is particularly useful for detecting small or partially visible objects such as cracks or damaged man-holes.

Tran et al. (2023) compared multiple YOLO-based models for pavement inspection tasks and found that single-stage detectors generally provide a good balance between accuracy and inference speed. This supports the use of YOLO-based detection in practical systems.

Li et al. (2023) explored anchor-free detection approaches for road defects in challenging environments with poor lighting. Their work showed improved robustness and reduced dependency on predefined anchor settings.

Other approaches such as transformer-based detectors and segmentation models have also been explored. However, these methods often require higher computational resources or dense annotations, which can limit their use in real-time systems.

### **2.1.2 Convolutional Neural Networks and Residual Architectures**

Convolutional neural networks (CNNs) are widely used for extracting visual features from images. Among them, residual networks (ResNets) remain a strong choice due to their ability to train deeper models effectively.

Wightman et al. (2021) showed that well-optimised ResNet models continue to perform competitively on image classification tasks.

He et al. (2022) demonstrated that self-supervised pre-training methods improve the generalisation ability of ResNet features, especially when labelled data is limited.

Nawaz et al. (2022) applied fine-tuned ResNet models to classify pavement cracks and reported high classification accuracy.

Although Vision Transformers have also been explored, they generally require larger datasets and higher computational resources. For moderate-sized datasets like road hazard datasets, CNN-based models remain more practical.

### **2.1.3 Infrastructure Monitoring Systems**

Recent advancements in sensing technologies and deep learning have enabled automated infrastructure monitoring systems.

Fan et al. (2022) proposed a real-time pothole detection system using stereo vision and deep learning.

Bahnsen et al. (2022) introduced a multi-modal system combining camera and LiDAR data.

Xu et al. (2019) developed a transformer-based crack detection model that provides accurate boundary segmentation.

These systems demonstrate the effectiveness of deep learning for road monitoring, while also highlighting computational and scalability challenges.

### **2.1.4 Quantum Machine Learning**

Quantum machine learning (QML) explores the use of quantum computing principles to enhance machine learning models.

Cerezo et al. (2022) analysed variational quantum algorithms and discussed challenges such as the barren plateau problem.

Schuld and Petruccione (2022) provided a theoretical foundation for quantum feature maps and kernel methods.

Biamonte (2021) demonstrated that variational quantum circuits can approximate general quantum operations.

Abbas et al. (2021) showed that quantum models can achieve higher representational capacity compared to classical models.

### **2.1.5 Hybrid Quantum-Classical Models**

Hybrid quantum-classical models combine classical neural networks with quantum circuits.

Kashif and Al-Kuwari (2023) studied cost function design in hybrid models.

Altares-López et al. (2021) applied hybrid models to image classification.

Huang et al. (2021) analysed the theoretical advantages of quantum-enhanced models.

Sagingalieva et al. (2023) applied hybrid models to real-world tasks with limited data.

These studies indicate that hybrid approaches can be effective for classification problems.

## **2.2 RESEARCH GAP**

From the literature review, it is evident that deep learning-based approaches perform well for road hazard detection. However, several limitations remain:

- Difficulty in generalising across different environments
- Class imbalance in real-world datasets
- Limited exploration of quantum-enhanced models in practical applications
- Lack of hybrid approaches applied specifically to road hazard classification

Additionally, most quantum machine learning studies are limited to benchmark datasets and do not address real-world infrastructure problems.

## **2.3 PROBLEM STATEMENT**

Road hazard detection and classification remain challenging due to variability in hazard appearance, similarities between different classes, and limitations in available datasets. Existing classical models may struggle to capture complex feature representations when dimensionality is reduced.

Therefore, there is a need for a robust and efficient system that can accurately classify road hazards while handling real-world variability. This work proposes a hybrid quantum-classical approach to address these challenges.

## **2.4 OBJECTIVES**

The objectives of this project are:

1. To design a two-stage system separating detection and classification
2. To develop a hybrid quantum-classical model using ResNet-18, MLP, and a Variational Quantum Circuit
3. To construct and preprocess a balanced dataset from multiple sources
4. To train the model using parameter-shift-based gradient computation
5. To evaluate performance using standard classification metrics and compare with a classical baseline

## **2.5 SUMMARY**

This chapter reviewed existing work in road hazard detection, deep learning, and quantum machine learning. Based on the identified research gaps, the problem statement and objectives of the proposed system were defined.

# CHAPTER 3

## TECHNICAL SPECIFICATION

### 3.1 REQUIREMENTS

#### 3.1.1 Functional Requirements

The functional requirements define the core operations that the proposed system must perform:

- The system should accept input images containing road surfaces.
- The system should detect regions of interest (hazards) using a YOLO-based detection model.
- The system should extract cropped regions corresponding to detected hazards.
- The system should preprocess images through resizing and normalisation.
- The system should classify hazards into one of four categories: crack, pothole, open manhole, or garbage.
- The system should integrate classical deep learning with a quantum circuit for classification.
- The system should output predicted class labels along with evaluation metrics.

#### 3.1.2 Non-Functional Requirements

The non-functional requirements define the quality attributes of the system:

- **Performance:** The system should provide accurate classification results with minimal latency.
- **Scalability:** The system should be capable of handling large datasets.
- **Reliability:** The system should produce consistent predictions under varying input conditions.

- **Maintainability:** The modular design should allow easy updates and improvements.
- **Usability:** The system should be simple to use for training and inference.

## 3.2 FEASIBILITY STUDY

### 3.2.1 Technical Feasibility

The system is technically feasible as it uses well-established frameworks such as PyTorch for deep learning and PennyLane for quantum simulation. The hybrid architecture can be implemented using existing tools without requiring specialised hardware.

The use of a quantum simulator allows experimentation with quantum models while avoiding limitations of current quantum hardware.

### 3.2.2 Economic Feasibility

The system is cost-effective as it relies on open-source tools and publicly available datasets. No expensive hardware is required since the quantum component is simulated.

This makes the project suitable for academic research and scalable for future deployment.

### 3.2.3 Operational Feasibility

The system can be integrated into existing road monitoring workflows. It can be used by municipal authorities or infrastructure management systems to automate hazard detection and classification.

The modular pipeline allows independent operation of detection and classification stages.

## 3.3 SYSTEM SPECIFICATION

### 3.3.1 Hardware Requirements

- Processor: Intel i5 or higher (or equivalent)
- GPU: NVIDIA GPU (optional but recommended for training)

- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 10 GB for datasets and models

### **3.3.2 Software Requirements**

- Operating System: Windows/Linux
- Programming Language: Python
- Deep Learning Framework: PyTorch
- Quantum Framework: PennyLane
- Libraries: NumPy, torchvision, matplotlib
- Development Environment: Jupyter Notebook / VS Code

## **3.4 SUMMARY**

This chapter described the technical specifications of the proposed system, including functional and non-functional requirements, feasibility analysis, and hardware and software specifications. These elements establish the foundation for the system design and implementation presented in the subsequent chapters.

# CHAPTER 4

## SYSTEM DESIGN

### 4.1 SYSTEM ARCHITECTURE

The proposed system follows a hybrid quantum-classical architecture designed to perform road hazard detection and classification in a modular manner. The architecture is divided into two major stages: detection and classification.

**Stage 1 — Detection:** A YOLO-based object detection model processes the input image and identifies potential hazard regions by generating bounding boxes.

**Stage 2 — Classification:** Each detected region is cropped and passed to a hybrid quantum-classical classifier, which predicts one of four hazard classes: crack, pothole, open manhole, or garbage.

This separation ensures modularity and allows independent optimisation of detection and classification components. During training, ground-truth annotations are used instead of predicted bounding boxes to avoid error propagation.

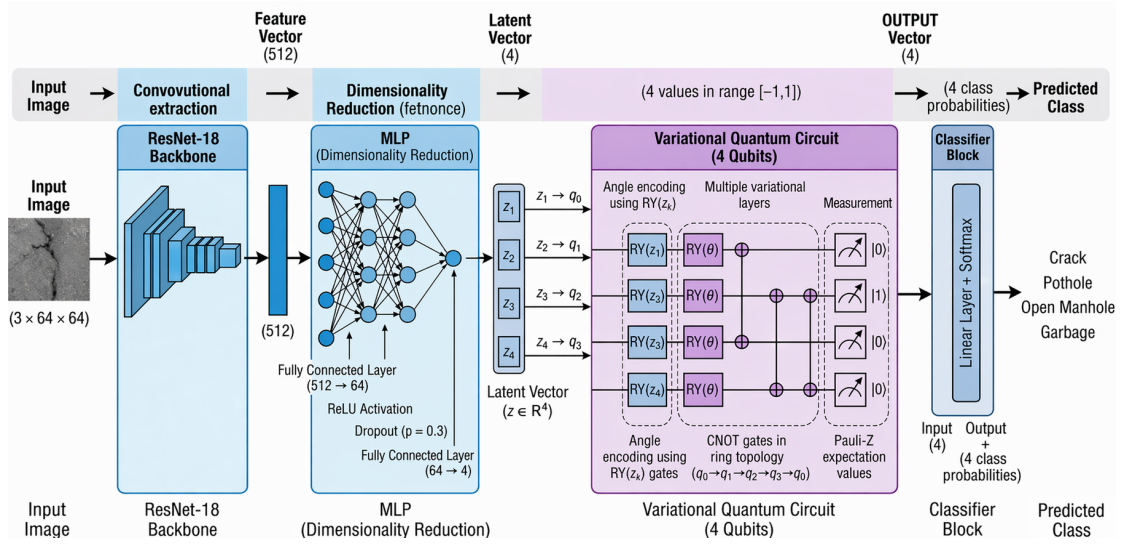


Figure 4.1: Overall architecture of the proposed quantum-classical framework

The architecture consists of the following components:

- Input image acquisition
- YOLO-based detection module

- Image preprocessing pipeline
- ResNet-18 feature extractor
- MLP dimensionality reduction layer
- Variational Quantum Circuit (VQC)
- Classical classification layer

## 4.2 DETAILED DESIGN

### 4.2.1 Input and Detection Module

The system takes a full-resolution image as input:

$$I \in \mathbb{R}^{H \times W \times 3} \quad (4.1)$$

The YOLO detector identifies regions of interest and outputs bounding boxes:

$$\mathbf{b} = (x_{\min}, y_{\min}, x_{\max}, y_{\max}) \quad (4.2)$$

These bounding boxes are used to extract hazard regions from the image.

### 4.2.2 Preprocessing Pipeline

The extracted regions are processed before being passed to the classifier.

**Cropping:**

$$I_{\text{crop}} = I[y_{\min} : y_{\max}, x_{\min} : x_{\max}] \quad (4.3)$$

**Resizing:**

$$I_{\text{resized}} = \mathcal{R}(I_{\text{crop}}, 64, 64) \quad (4.4)$$

**Normalisation:**

$$\hat{x}_c = \frac{x_c - \mu_c}{\sigma_c} \quad (4.5)$$

This ensures compatibility with the pre-trained ResNet model.

### 4.2.3 ResNet-18 Backbone

The normalised image is passed through a pre-trained ResNet-18 model for feature extraction.

Residual learning is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (4.6)$$

When dimensions differ:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + W_s \mathbf{x} \quad (4.7)$$

Global average pooling produces:

$$\mathbf{f} \in \mathbb{R}^{512} \quad (4.8)$$

This feature vector captures high-level representations of the hazard.

### 4.2.4 MLP Dimensionality Reduction

The feature vector is reduced to match the number of qubits.

$$\mathbf{h} = \text{ReLU}(W_1 \mathbf{f} + \mathbf{b}_1) \quad (4.9)$$

$$\mathbf{z} = W_2 \mathbf{h} + \mathbf{b}_2 \quad (4.10)$$

The output vector:

$$\mathbf{z} \in \mathbb{R}^4 \quad (4.11)$$

This dimensionality reduction is essential for quantum encoding.

### 4.2.5 Variational Quantum Circuit

The reduced feature vector is encoded into a 4-qubit quantum system.

### Angle Encoding

$$|\psi_{\text{enc}}\rangle = \bigotimes_{k=1}^4 R_Y(z_k)|0\rangle \quad (4.12)$$

### Variational Layers

$$|\psi_{\text{out}}\rangle = \prod_{\ell=1}^L U_{\ell}(\boldsymbol{\theta}^{\ell})|\psi_{\text{enc}}\rangle \quad (4.13)$$

Each layer consists of:

- Parameterised  $R_Y$  rotations
- CNOT gates in ring topology

### Measurement

$$e_k = \langle \psi_{\text{out}} | Z_k | \psi_{\text{out}} \rangle \quad (4.14)$$

The output vector:

$$\mathbf{e} \in [-1, 1]^4 \quad (4.15)$$

#### 4.2.6 Classification Layer

The quantum output is mapped to class predictions:

$$\hat{\mathbf{y}} = W_{\text{cls}} \mathbf{e} + \mathbf{b}_{\text{cls}} \quad (4.16)$$

$$P(y = i) = \frac{\exp(\hat{y}_i)}{\sum_j \exp(\hat{y}_j)} \quad (4.17)$$

$$\hat{c} = \arg \max_i P(y = i) \quad (4.18)$$

#### 4.2.7 System Flow

The complete system flow can be summarised as:

1. Input image is processed by YOLO detector

2. Hazard regions are extracted
3. Images are resized and normalised
4. Features are extracted using ResNet-18
5. Features are reduced using MLP
6. Latent vector is encoded into quantum circuit
7. Variational layers process quantum state
8. Measurement produces classical vector
9. Final classification is performed using softmax

### **4.3 SUMMARY**

This chapter presented the system design of the proposed hybrid quantum-classical framework. The architecture integrates deep learning and quantum computing components in a structured pipeline, enabling efficient and accurate road hazard classification.

# CHAPTER 5

## METHODOLOGY AND TESTING

### 5.1 METHODOLOGY

The proposed system follows a hybrid quantum-classical pipeline for road hazard classification. The methodology is structured as a two-stage process consisting of detection and classification.

#### 5.1.1 Stage 1: Detection

The input image is represented as:

$$I \in \mathbb{R}^{H \times W \times 3} \quad (5.1)$$

A YOLO-based detector processes the image and generates bounding boxes:

$$\mathbf{b}_i = (x_{\min}, y_{\min}, x_{\max}, y_{\max}) \quad (5.2)$$

These bounding boxes define regions of interest corresponding to potential road hazards.

#### 5.1.2 Stage 2: Hybrid Classification Model

Each detected region is cropped and passed to the hybrid quantum-classical classifier.

##### Feature Extraction

The cropped and preprocessed image is passed through a ResNet-18 backbone. Residual learning is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (5.3)$$

Global average pooling produces:

$$\mathbf{f} \in \mathbb{R}^{512} \quad (5.4)$$

## Dimensionality Reduction

The feature vector is reduced using a two-layer MLP:

$$\mathbf{h} = \text{ReLU}(W_1 \mathbf{f} + \mathbf{b}_1) \quad (5.5)$$

$$\mathbf{z} = W_2 \mathbf{h} + \mathbf{b}_2 \quad (5.6)$$

$$\mathbf{z} \in \mathbb{R}^4 \quad (5.7)$$

## Quantum Processing

**Angle Encoding:**

$$|\psi_{\text{enc}}\rangle = \bigotimes_{k=1}^4 R_Y(z_k) |0\rangle \quad (5.8)$$

**Variational Layers:**

$$|\psi_{\text{out}}\rangle = \prod_{\ell=1}^L U_{\ell}(\theta^{\ell}) |\psi_{\text{enc}}\rangle \quad (5.9)$$

**Measurement:**

$$e_k = \langle \psi_{\text{out}} | Z_k | \psi_{\text{out}} \rangle \quad (5.10)$$

## Classification

$$\hat{\mathbf{y}} = W_{\text{cls}} \mathbf{e} + \mathbf{b}_{\text{cls}} \quad (5.11)$$

$$P(y = i) = \frac{\exp(\hat{y}_i)}{\sum_j \exp(\hat{y}_j)} \quad (5.12)$$

$$\hat{c} = \arg \max_i P(y = i) \quad (5.13)$$

---

**Algorithm 1** Quantum-Enhanced Road Hazard Classification Pipeline

---

- 1: **Input:** Image  $I \in \mathbb{R}^{H \times W \times 3}$
  - 2: **Output:** Predicted class label  $\hat{c} \in \{1, 2, 3, 4\}$
  - 3: Detect bounding box  $\mathbf{b}$  using YOLO
  - 4: Crop region:  $I_{\text{crop}} = I[y_{\min} : y_{\max}, x_{\min} : x_{\max}]$
  - 5: Resize to  $64 \times 64$
  - 6: Normalize image
  - 7: Extract features using ResNet-18
  - 8: Reduce dimension using MLP
  - 9: Encode into quantum circuit
  - 10: Apply variational quantum layers
  - 11: Measure expectation values
  - 12: Compute logits and apply softmax
  - 13: Output predicted class
- 

## 5.2 ALGORITHM

## 5.3 TESTING PROCEDURE

### 5.3.1 Training Configuration

The model is trained using supervised learning with the following configuration:

- Optimiser: Adam
- Learning rate:  $1 \times 10^{-4}$
- Batch size: 32
- Epochs: 50
- Weight decay:  $1 \times 10^{-4}$

### 5.3.2 Loss Function

The model uses categorical cross-entropy loss:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log P(y = c) \quad (5.14)$$

### 5.3.3 Quantum Gradient Computation

Gradients for the quantum circuit are computed using the parameter-shift rule:

$$\frac{\partial \langle O \rangle}{\partial \theta_k} = \frac{1}{2} \left[ \langle O \rangle_{\theta_k + \frac{\pi}{2}} - \langle O \rangle_{\theta_k - \frac{\pi}{2}} \right] \quad (5.15)$$

#### 5.3.4 Evaluation Metrics

The model is evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

#### 5.4 SUMMARY

This chapter described the methodology and testing procedures of the proposed system. The hybrid architecture integrates classical feature extraction with quantum processing, and the model is evaluated using standard performance metrics.

# CHAPTER 6

## PROJECT IMPLEMENTATION

### 6.1 IMPLEMENTATION OVERVIEW

This chapter describes the implementation details of the proposed hybrid quantum-classical road hazard classification system. It includes the software environment, training configuration, hyperparameters, experimental setups, and evaluation methodology used during model development.

### 6.2 DEVELOPMENT ENVIRONMENT

#### 6.2.1 Software Tools

The system is implemented using the following software stack:

- **Programming Language:** Python
- **Deep Learning Framework:** PyTorch (version 2.x)
- **Quantum Framework:** PennyLane with `default.qubit` simulator
- **Object Detection:** YOLO (Ultralytics implementation)
- **Libraries:** NumPy, torchvision, matplotlib
- **Development Tools:** Collab Notebook

PennyLane integrates with PyTorch through automatic differentiation, enabling seamless hybrid model training.

#### 6.2.2 Hardware Setup

Experiments are conducted on a system with GPU acceleration (if available), allowing efficient training of deep learning models and simulation of quantum circuits.

## 6.3 TRAINING CONFIGURATION

### 6.3.1 Loss Function

The model is trained using categorical cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log P(y = c \mid \mathbf{e}^{(i)}) \quad (6.1)$$

### 6.3.2 Optimisation

All parameters, including the ResNet backbone, MLP layers, quantum circuit parameters, and classification head, are optimised using the Adam optimiser:

$$\phi_{t+1} = \phi_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (6.2)$$

### 6.3.3 Quantum Gradient Computation

Gradients for quantum parameters are computed using the parameter-shift rule:

$$\frac{\partial \langle O \rangle}{\partial \theta_k} = \frac{1}{2} \left[ \langle O \rangle_{\theta_k + \frac{\pi}{2}} - \langle O \rangle_{\theta_k - \frac{\pi}{2}} \right] \quad (6.3)$$

PennyLane internally manages gradient computation and integrates it with PyTorch's automatic differentiation.

## 6.4 HYPERPARAMETERS

The training hyperparameters used in the implementation are listed in Table 6.1.

## 6.5 EXPERIMENTAL CONFIGURATIONS

To evaluate the effectiveness of the hybrid model, four configurations are tested:

1. Classical dataset with classical model
2. Classical dataset with quantum model
3. Augmented dataset with classical model

Table 6.1: Training Hyperparameters

| Hyperparameter             | Value                   |
|----------------------------|-------------------------|
| Learning rate ( $\alpha$ ) | $1 \times 10^{-4}$      |
| Batch size ( $B$ )         | 32                      |
| Epochs ( $E$ )             | 50                      |
| Weight decay ( $\lambda$ ) | $1 \times 10^{-4}$      |
| VQC depth ( $L$ )          | 6                       |
| Number of qubits ( $n$ )   | 4                       |
| Dropout rate               | 0.3                     |
| Optimiser                  | Adam                    |
| Loss function              | Cross-entropy           |
| Quantum backend            | PennyLane default.qubit |

#### 4. Augmented dataset with quantum model

This setup enables comparison between classical and quantum approaches.

## 6.6 EVALUATION METRICS

Model performance is evaluated using standard classification metrics.

### 6.6.1 Accuracy

$$\text{Accuracy} = \frac{\sum_{c=1}^C \text{TP}_c}{\sum_{c=1}^C (\text{TP}_c + \text{FP}_c + \text{FN}_c + \text{TN}_c)} \quad (6.4)$$

### 6.6.2 Precision

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \quad (6.5)$$

### 6.6.3 Recall

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (6.6)$$

### 6.6.4 F1-Score

$$\text{F1}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (6.7)$$

$$\overline{\text{F1}} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c \quad (6.8)$$

### 6.6.5 Confusion Matrix

The confusion matrix provides a detailed class-wise performance analysis.

### 6.6.6 mAP@50

For detection performance, mean Average Precision at IoU threshold 0.50 is used.

## 6.7 BASELINE MODEL

A classical baseline is implemented by replacing the quantum circuit with a linear layer:

$$\hat{\mathbf{y}} = W_{\text{cls}}\mathbf{z} + \mathbf{b}_{\text{cls}} \quad (6.9)$$

This allows direct comparison between classical and hybrid models.

## 6.8 SUMMARY

This chapter presented the implementation details of the proposed system, including development environment, training configuration, hyperparameters, and evaluation methodology. The results obtained using this setup are discussed in the next chapter.

# CHAPTER 7

## RESULTS AND DISCUSSION

### 7.1 INTRODUCTION

This chapter presents the experimental results obtained from the proposed quantum-enhanced road hazard classification framework. The results are analysed in terms of overall performance, per-class behaviour, qualitative outputs, intermediate representations, and training convergence.

### 7.2 OVERALL CLASSIFICATION PERFORMANCE

Table 7.1 presents the overall classification performance on the test set for all four experimental configurations.

Table 7.1: Performance Comparison Across Dataset and Model Configurations

| Configuration                            | mAP@50        | Precision     | Recall        | F1            |
|------------------------------------------|---------------|---------------|---------------|---------------|
| Quantum Dataset + Classical Model        | 0.3150        | 0.6203        | 0.3056        | 0.4094        |
| Quantum Dataset + Quantum Model          | 0.0016        | 0.3366        | 0.5802        | 0.4260        |
| Classical Dataset + Classical Model      | 0.8032        | 0.8503        | 0.7581        | 0.8016        |
| <b>Classical Dataset + Quantum Model</b> | <b>0.6788</b> | <b>0.8579</b> | <b>0.8451</b> | <b>0.8439</b> |

#### 7.2.1 Analysis of Experimental Configurations

The results highlight several key observations.

**Impact of Dataset Quality:** Configurations using the curated and balanced dataset achieve significantly better performance than those using the alternative dataset. This shows that dataset quality plays a dominant role in classification performance.

**Hybrid Model Improvement:** The hybrid model achieves an F1-score of 0.8439, compared to 0.8016 for the classical baseline. This indicates that the quantum component contributes to improved performance.

**Recall Advantage:** A notable improvement is observed in recall (0.8451 vs. 0.7581), which is important in road hazard detection, where missed detections can have safety implications.

**Quantum Dataset Performance:** The poor performance of configurations using the alternative dataset suggests that the preprocessing approach used in those experiments is not suitable for reliable classification.

### 7.3 PER-CLASS ANALYSIS

Table 7.2 reports per-class precision, recall, and F1-score for the proposed hybrid model.

Table 7.2: Per-Class Metrics — Proposed Hybrid Model (Test Set)

| Class                | Precision   | Recall      | F1-Score    |
|----------------------|-------------|-------------|-------------|
| Pothole              | 0.94        | 0.91        | 0.92        |
| Crack                | 0.92        | 0.73        | 0.82        |
| Garbage              | 0.79        | 0.98        | 0.88        |
| Open Manhole         | 0.74        | 0.85        | 0.79        |
| <b>Macro Average</b> | <b>0.85</b> | <b>0.87</b> | <b>0.85</b> |

#### 7.3.1 Per-Class Discussion

**Pothole:** The model achieves high precision and recall, indicating that potholes are well distinguished due to their clear visual characteristics.

**Crack:** Crack detection shows high precision but lower recall, suggesting that some crack instances are missed due to variability in appearance.

**Garbage:** High recall but lower precision indicates that the model occasionally over-predicts this class.

**Open Manhole:** Moderate performance suggests confusion with similar circular or shadow patterns.

### 7.4 QUALITATIVE PIPELINE VISUALISATION

The pipeline successfully detects and classifies hazard regions, demonstrating the end-to-end functionality of the system.

### 7.5 INTERMEDIATE FEATURE REPRESENTATIONS

#### 7.5.1 ResNet Features

The ResNet backbone extracts discriminative features capturing texture and structural characteristics of hazards.



Figure 7.1: Sample input image from the test set

### Cropped Image



Figure 7.2: YOLO-based cropped hazard region

#### **7.5.2 MLP Latent Space**

The MLP reduces dimensionality while preserving class separability.

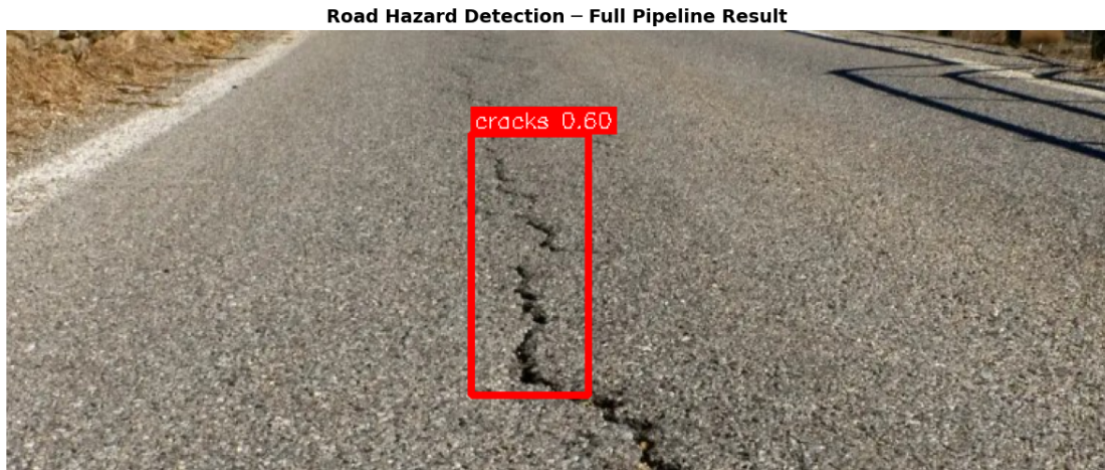


Figure 7.3: Final output of the pipeline with predicted class

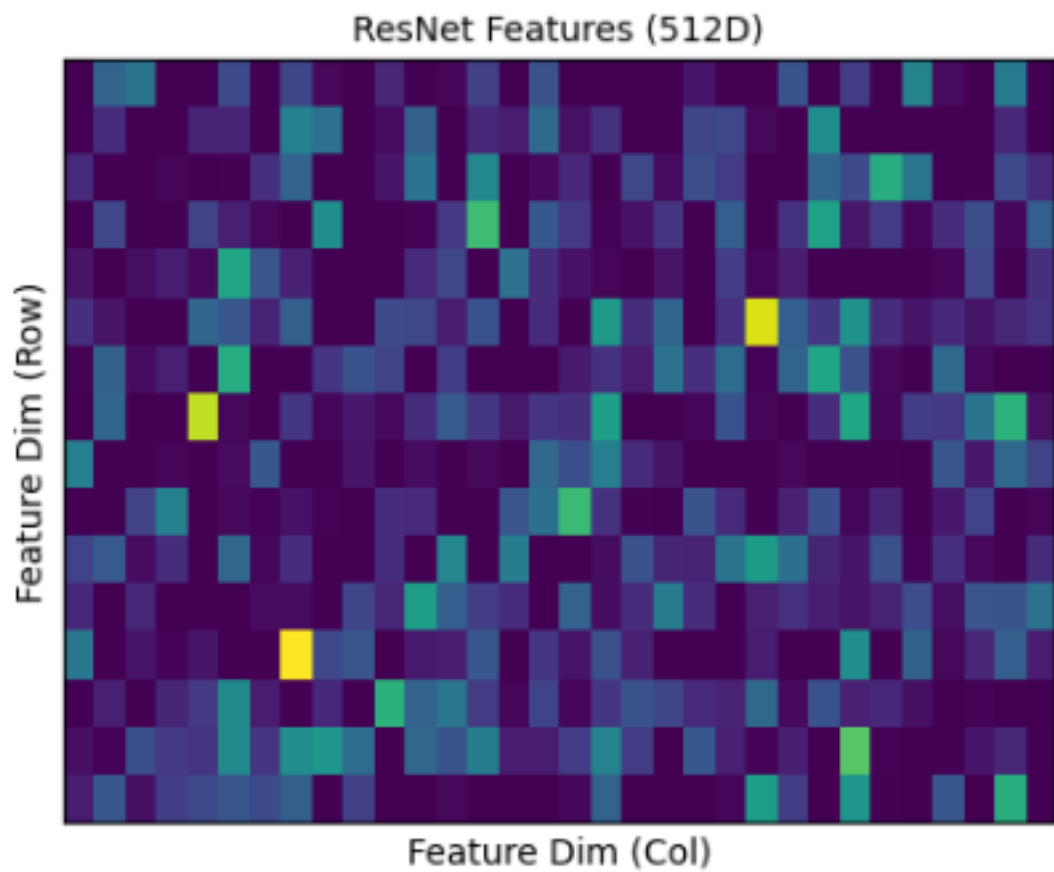


Figure 7.4: ResNet-18 feature representations

### 7.5.3 Quantum Measurements

The VQC produces distinct measurement patterns across classes, indicating meaningful transformation.

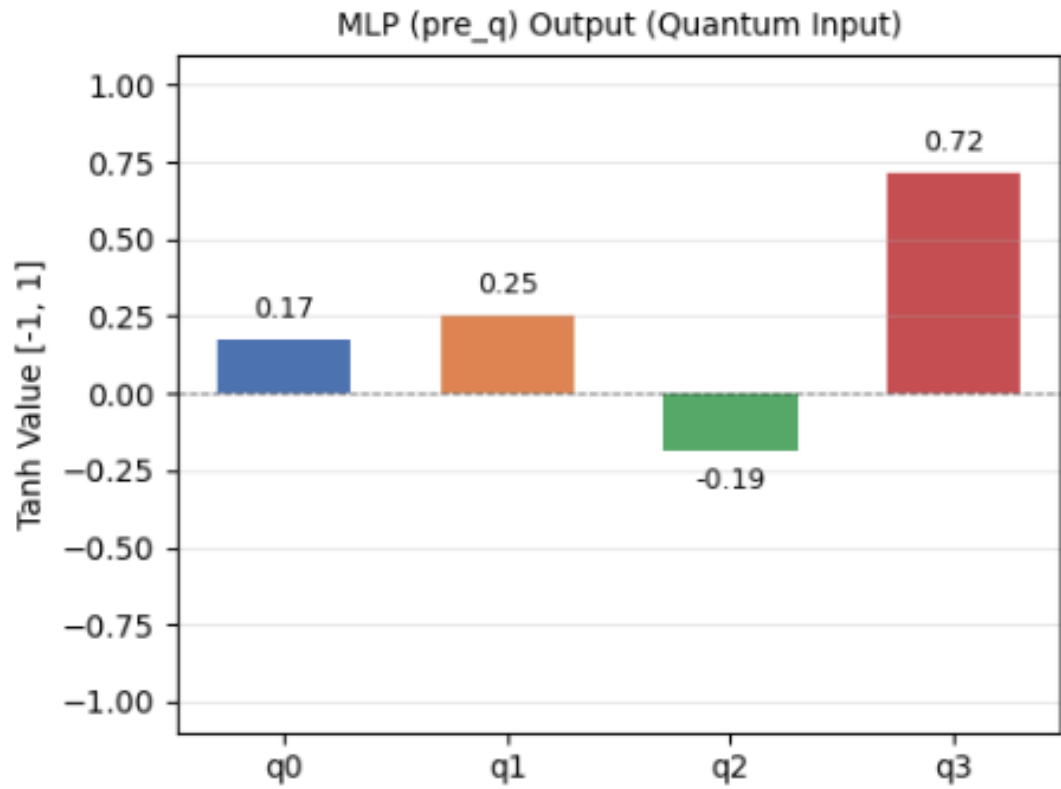


Figure 7.5: MLP representation

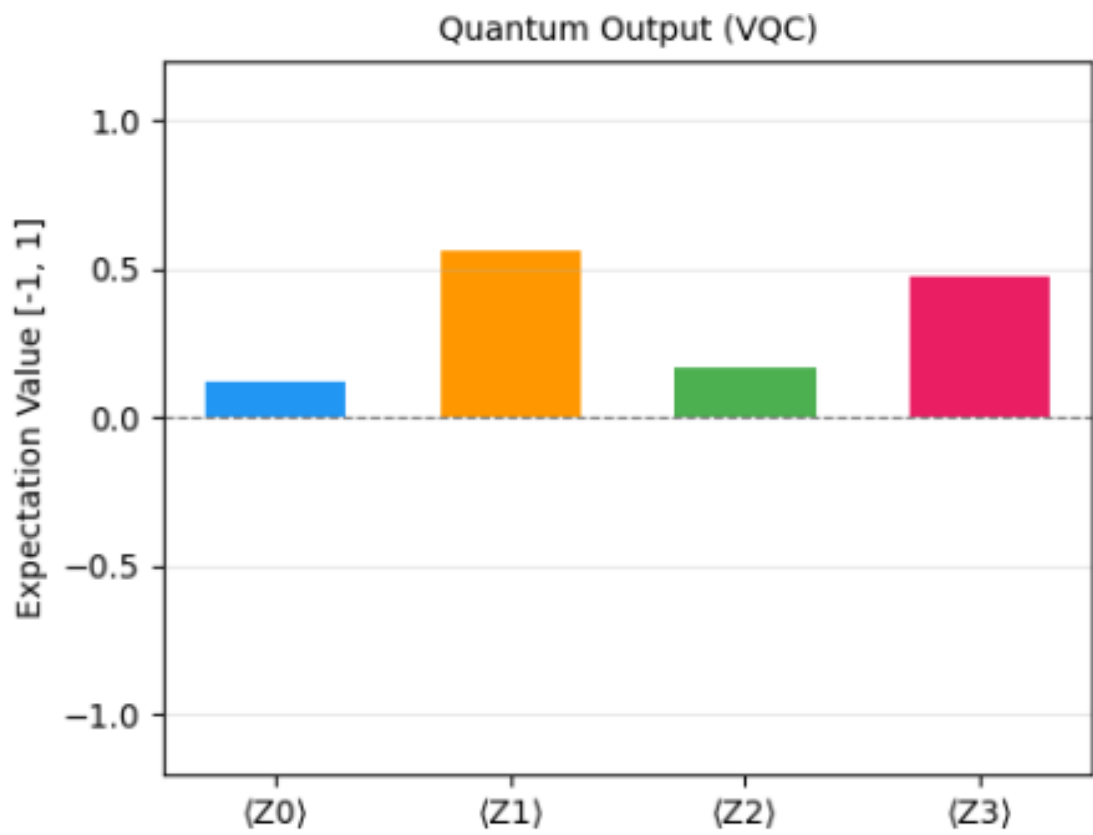


Figure 7.6: VQC measurement outputs across classes

## 7.6 TRAINING CONVERGENCE ANALYSIS

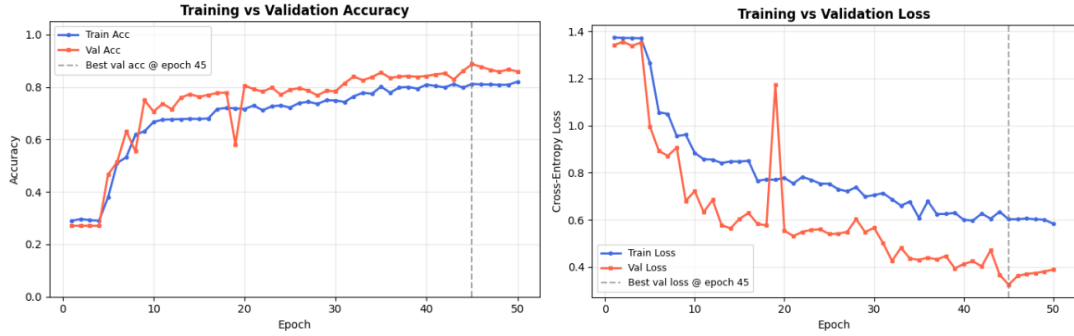


Figure 7.7: Training and validation accuracy (left) and loss (right)

### 7.6.1 Observations

**Stable Convergence:** The model converges smoothly without instability.

**Peak Performance:** Best validation accuracy is observed around epoch 45.

**Loss Behaviour:** Training loss decreases consistently, while validation loss shows minor fluctuations.

## 7.7 DISCUSSION

### 7.7.1 Hybrid Model Effectiveness

The hybrid model demonstrates improved performance compared to the classical baseline, indicating that quantum components can contribute useful representations.

### 7.7.2 Role of VQC

The VQC introduces additional transformation capability beyond a classical linear layer, contributing to improved classification performance.

### 7.7.3 Limitations

- The quantum circuit is simulated rather than executed on real hardware.
- Input resolution is limited to  $64 \times 64$ .
- Dataset diversity may not fully represent real-world conditions.

## **7.8 SUMMARY**

The hybrid model achieves strong performance, with improved recall and balanced classification across hazard types. The results confirm that combining classical deep learning with quantum circuits is a promising approach for road hazard classification.

## CHAPTER 8

### CONCLUSION AND FURTHER ENHANCEMENTS

#### 8.1 CONCLUSION

This thesis presented the design, implementation, and evaluation of a quantum-enhanced deep learning framework for road hazard safety analysis. The proposed system addresses the problem of automated road hazard detection and classification, which is important for urban infrastructure safety and smart city applications. A hybrid approach combining classical deep learning and quantum computing techniques was developed within a modular two-stage pipeline.

##### 8.1.1 Summary of Contributions

The main contributions of this work are summarised below:

**1. Modular Two-Stage Pipeline:** A two-stage pipeline was designed, separating road hazard detection (YOLO-based) from classification (hybrid quantum-classical model). This modular design allows independent optimisation of each stage and makes the system easier to extend.

**2. Hybrid Quantum-Classical Classifier:** A hybrid model was developed consisting of:

- A pre-trained ResNet-18 backbone producing a 512-dimensional feature vector from  $64 \times 64$  images.
- A two-layer MLP ( $512 \rightarrow 64 \rightarrow 4$ ) for dimensionality reduction.
- A 4-qubit Variational Quantum Circuit with  $R_Y$  encoding, 6 variational layers, and CNOT-based entanglement.
- A classical linear layer with softmax output for final classification.

**3. Class-Balanced Multi-Source Dataset:** A dataset was constructed by merging two publicly available datasets and applying class balancing through downsampling. The final dataset includes 3,340 training, 740 validation, and 258 test images across four hazard classes.

**4. End-to-End Hybrid Training:** The model was trained end-to-end using the Adam optimiser. Classical gradients were computed using automatic differentiation, while quantum gradients were obtained using the parameter-shift rule.

**5. Comparative Evaluation:** The hybrid model achieved a macro F1-score of 0.8439, outperforming the classical baseline by approximately 5%. The results demonstrate that the quantum component contributes to improved classification performance.

### 8.1.2 Key Findings

The experimental results lead to the following observations:

1. The hybrid quantum-classical model is a viable approach for road hazard classification and performs competitively compared to a classical baseline.
2. Dataset quality and class balance have a strong influence on performance, often more significant than model complexity.
3. The quantum component improves recall, which is particularly important for safety-critical applications.
4. The hybrid model trains stably, and the parameter-shift rule provides reliable gradient estimation.
5. The modular design simplifies system development and allows independent improvements to detection and classification stages.

### 8.1.3 Significance of the Work

This work demonstrates the practical applicability of hybrid quantum-classical models in a real-world computer vision task. It provides a proof of concept for integrating variational quantum circuits into infrastructure monitoring systems.

The proposed framework also contributes to smart city applications by enabling automated road hazard detection, which can support maintenance planning and improve road safety.

## 8.2 LIMITATIONS

Despite promising results, the current work has several limitations:

1. **Quantum Simulation:** The quantum circuit is executed on a simulator rather than real hardware. Performance may differ when deployed on noisy quantum devices.
2. **Low Resolution Input:** The use of  $64 \times 64$  images may limit the model's ability to capture fine-grained details.
3. **Limited Qubit Count:** The model uses only 4 qubits, which restricts representational capacity.
4. **Dataset Scope:** The dataset includes only four hazard categories and limited geographic diversity.
5. **Barren Plateau Risk:** Deeper quantum circuits may suffer from vanishing gradients, making training difficult.

## 8.3 FURTHER ENHANCEMENTS

Several directions can be explored to extend this work:

### 8.3.1 Deployment on Quantum Hardware

Future work should evaluate the model on real quantum devices (e.g., IBM Quantum or IonQ) to study the effect of noise and hardware constraints.

### 8.3.2 Noise-Robust Circuit Design

Developing hardware-efficient and noise-tolerant quantum circuits can improve real-world performance.

### 8.3.3 Scaling Qubit Count

Increasing the number of qubits and latent dimensions may enhance model expressibility and classification accuracy.

### 8.3.4 Severity-Based Classification

Extending the model to classify hazard severity (e.g., minor vs severe potholes) would provide more actionable insights.

### **8.3.5 Real-Time System Integration**

Integrating the framework with real-time systems such as vehicle-mounted cameras or surveillance networks can enable continuous monitoring.

### **8.3.6 Larger and Diverse Datasets**

Expanding the dataset with more samples and varied environments will improve generalisation.

### **8.3.7 Quantum Kernel Methods**

Alternative approaches such as quantum kernel methods could be explored for comparison.

### **8.3.8 Explainability**

Developing interpretability techniques for both classical and quantum components will improve trust and usability.

## **8.4 CLOSING REMARKS**

This thesis demonstrates that hybrid quantum-classical learning is a feasible approach for road hazard classification. By combining deep learning with variational quantum circuits, the proposed system achieves strong performance while laying the foundation for future research in quantum-enhanced computer vision systems.

## BIBLIOGRAPHY

- Abbas, A., Sutter, D., Zoufal, C., Lucchi, A., Figalli, A., and Woerner, S. (2021). The power of quantum neural networks. *Nature Computational Science*, 1(6):403–409.
- Altares-López, S., Ribeiro, A., and García-Ripoll, J. J. (2021). Automatic design of quantum feature maps. *Quantum Science and Technology*, 6(4):045015.
- Bahnsen, C. K., Møgelmoose, A., and Moeslund, T. B. (2022). Rain and snow removal in image sequences using multiple temporal and spatial cues. *IEEE Transactions on Intelligent Vehicles*, 7(2):476–488.
- Biamonte, J. (2021). Universal variational quantum computation. *Physical Review A*, 103(3):L030401.
- Cerezo, M., Verdon, G., Huang, H.-Y., Cincio, L., and Coles, P. J. (2022). Challenges and opportunities of near-term quantum computing systems. *Proceedings of the IEEE*, 110(6):888–907.
- Fan, R., Wang, H., Cai, P., and Liu, M. (2022). SNE-RoadSeg+: Customer-centric road surface normal estimation for autonomous driving. *IEEE Transactions on Intelligent Transportation Systems*, 23(7):9033–9043.
- He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. (2022). Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009.
- Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., and McClean, J. R. (2021). Power of data in quantum machine learning. *Nature Communications*, 12:2631.
- Kashif, M. and Al-Kuwari, S. (2023). The impact of cost function globality and locality in hybrid quantum-classical neural networks on NISQ devices. *Machine Learning: Science and Technology*, 4(1):015004.

- Li, Y., Zhao, J., and Wang, Q. (2023). Road defect detection in complex scenes using an improved anchor-free detector. *Engineering Applications of Artificial Intelligence*, 126:106876.
- Nawaz, T., Mansoor, F., and Masood, S. (2022). Automated pavement crack severity classification using fine-tuned convolutional neural networks. *Automation in Construction*, 142:104483.
- Sagingalieva, A., Kordzanganeh, M., Kenbayev, N., Kosichkina, D., Tomashuk, T., and Melnikov, A. (2023). Hybrid quantum neural network for drug response prediction. *Cancers*, 15(10):2705.
- Schuld, M. and Petruccione, F. (2022). *Machine Learning with Quantum Computers*. Springer, Cham, 2nd edition.
- Tran, V., Nguyen, T., and Le, H. (2023). Pavement distress detection using deep learning: A systematic review. *Construction and Building Materials*, 368:130302.
- Wang, C.-Y., Bochkovskiy, A., and Liao, H.-Y. M. (2023). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7464–7475.
- Wang, C.-Y., Yeh, I.-H., and Liao, H.-Y. M. (2024). YOLOv9: Learning what you want to learn using programmable gradient information. *arXiv preprint arXiv:2402.13616*.
- Wightman, R., Touvron, H., and Jégou, H. (2021). ResNet strikes back: An improved baseline in 2.7 years of research. *arXiv preprint arXiv:2110.00476*.
- Xu, H., Su, X., Wang, Y., Cai, H., Cui, K., and Chen, X. (2019). Automatic bridge crack detection using a convolutional neural network. *Applied Sciences*, 9(14):2867.

## APPENDIX A: SOURCE CODE

The source code for the proposed quantum-enhanced road hazard classification framework is presented in this appendix. The implementation combines PyTorch for classical deep learning components and PennyLane for quantum circuit simulation.

### A.1 HYBRID QUANTUM-CLASSICAL MODEL

---

```
import torch
import torch.nn as nn
import pennylane as qml
from torchvision import models

# Quantum device configuration
n_qubits = 4
n_layers = 6
dev = qml.device("default.qubit", wires=n_qubits)

@qml.qnode(dev, interface="torch", diff_method="parameter-shift")
def quantum_circuit(inputs, weights):

    # Angle encoding
    for i in range(n_qubits):
        qml.RY(inputs[i], wires=i)

    # Variational layers
    for layer in range(n_layers):

        # Parameterised rotations
        for i in range(n_qubits):
            qml.RY(weights[layer, i], wires=i)

        # Ring entanglement
        for i in range(n_qubits):
            qml.CNOT(wires=[i, (i + 1) % n_qubits])

    # Measurement
    return [qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]

class HybridQuantumClassifier(nn.Module):

    def __init__(self, num_classes=4):
```

```

super().__init__()

# ResNet-18 feature extractor
resnet = models.resnet18(pretrained=True)
self.backbone = nn.Sequential(*list(resnet.children())[:-1])

# MLP dimensionality reduction
self.mlp = nn.Sequential(
    nn.Linear(512, 64),
    nn.ReLU(),
    nn.Dropout(0.3),
    nn.Linear(64, n_qubits)
)

# Quantum layer
weight_shapes = {"weights": (n_layers, n_qubits)}
self.vqc = qml.qnn.TorchLayer(quantum_circuit, weight_shapes)

# Classification head
self.classifier = nn.Linear(n_qubits, num_classes)

def forward(self, x):

    # Extract features
    f = self.backbone(x).flatten(1)

    # Reduce dimension
    z = self.mlp(f)

    # Apply quantum circuit per sample
    q_out = torch.stack([self.vqc(z[i]) for i in range(z.size(0))])

    # Final classification
    logits = self.classifier(q_out)

    return logits

```

---

Listing 8.1: Hybrid quantum-classical model with ResNet-18 backbone, MLP, and 4-qubit VQC.

---

## A.2 DATA PREPROCESSING PIPELINE

---

```

from torchvision import transforms

transform = transforms.Compose([

```

```

        transforms.Resize((64, 64)),
        transforms.ToTensor(),
        transforms.Normalize(
            mean=[0.485, 0.456, 0.406],
            std=[0.229, 0.224, 0.225]
        )
    ])

```

---

Listing 8.2: Image preprocessing pipeline including resizing and normalisation.

—

## A.3 TRAINING LOOP

---

```

import torch.optim as optim

model = HybridQuantumClassifier(num_classes=4)

optimizer = optim.Adam(
    model.parameters(),
    lr=1e-4,
    weight_decay=1e-4
)

criterion = nn.CrossEntropyLoss()

num_epochs = 50

for epoch in range(num_epochs):

    model.train()

    for images, labels in train_loader:

        optimizer.zero_grad()

        outputs = model(images)

        loss = criterion(outputs, labels)

        loss.backward()

        optimizer.step()

    print(f"Epoch {epoch+1}/{num_epochs}, Loss: {loss.item():.4f}")

```

---

Listing 8.3: Training loop using Adam optimiser and cross-entropy loss.

---

## A.4 INFERENCE PIPELINE

---

```
model.eval()

with torch.no_grad():

    for images, labels in test_loader:

        outputs = model(images)

        _, preds = torch.max(outputs, 1)
```

---

Listing 8.4: Inference pipeline for prediction on test images.

## **APPENDIX B: PUBLICATION**

Navanith D, Sree Ramkumar M, V Hemal Sri, and P Antony Seba. *Quantum-Enhanced Deep Learning Framework for Road Hazard Safety Analysis*. Vellore Institute of Technology, Vellore, 2024.